

Large gap control in electromagnetic levitation

Subrata Banerjee*

Department of Electrical Engineering, National Institute of Technology, Durgapur-713209, India

Dinkar Prasad

Department of Electrical Engineering, Indian Institute of Technology, Kharagpur-721302, India

Jayanta Pal

Department of Electrical Engineering, Indian Institute of Technology, Kharagpur-721302, India

(Received 29 April 2005; accepted 11 December 2005)

Abstract

This paper describes design and implementation of a single axis dc attraction type electromagnetic suspension system where an electromagnet of 2.6 kg mass is levitated over a large gap under a fixed ferromagnetic guide-way. The electromagnet exhibits nonlinear force-current-distance characteristics, and if controllers are to be designed by using linear analysis, the air-gap is restricted to a small region around the chosen nominal operating point. In this work, an attempt has been made to increase the operating range of an electromagnetic suspension system by using the concept of piecewise linear control where the nonlinear force-current-airgap relationships of the magnetic suspension system have been successively linearized at several operating points with a suitable controller designed for each operating point. A novel analog switching scheme has been designed and implemented to automatically switch to the relevant controller depending on the actual air-gap. © 2006 ISA—The Instrumentation, Systems, and Automation Society.

Keywords: Electromagnetic levitation; Piecewise linear control; Analog switching; Switched mode power circuit

1. Introduction

The use of magnetic force to suspend objects with no visible means of support is termed magnetic levitation or MAGLEV. Based on the principle of operation, magnetic levitation may broadly be classified into two types: electrodynamic levitation and electromagnetic levitation. Electrodynamical systems actuate through repulsive forces and normally utilize superconducting magnets to generate the forces. In electromagnetic systems, levitation is produced due to the attractive

forces between the electromagnets and ferromagnetic objects. Modern applications of levitation in equipments like magnetic bearings and magnetically levitated vehicles have given renewed impetus to research efforts in this area. Some other applications of electromagnetic levitation include levitation of models in a wind tunnel, vibration isolation of sensitive machinery, levitation of molten metal in induction furnaces, levitation of metal slabs during manufacture, etc.

The electromagnetic levitation system (EMLS) is inherently unstable and strongly nonlinear in nature [1]. For obtaining closed-loop stability, controllers [1,2] have been designed based on a model obtained by linearizing at a nominal operating gap.

*Tel.: 91 0343 2544804; E-mail address:
bansub2004@rediffmail.com

It has been found that such controllers provide satisfactory performance in the neighborhood of the operating point and performance degradation occurs as the operating point is changed. It becomes difficult for the controller to maintain stability if there is some variation in the operating gap. Due to the inherent nonlinear nature of the system, the uncertainty in the nominal linearized model increases as one moves away from the nominal operating point, and, as a result, the controller becomes practically invalid for large variations in the operating gap. But from an implementation point of view there is an urgent need to enhance the capability of the controller so that the object can stably levitate over a large variation in air-gap. A solution that has been reported in the literature is to design a nonlinear controller by considering the nonlinear model of the system, and, normally, a feedback-linearizing controller [3] is chosen. Normally, finding the exact nonlinear model of the system, and design and implementation of the controller, is not an easy task.

Most real-life physical processes are actually nonlinear systems. Conventional design approaches use different approximation methods to handle the nonlinearity. Some typical choices are linear, piecewise linear, and look-up table approximations to trade off factors of complexity, cost, and system performance. In this work, an attempt has been made to solve this problem by using the concept of piecewise linear control. Piecewise linear control is often used in a wide range of engineering systems. An inherently nonlinear system operating at different operating points or subject to physical constraints may be modeled as a piecewise linear system. A piecewise linear controller is relatively simple and easy to implement through analog means. However, in certain applications, it may limit control performance and may be costly to implement. A piecewise linear controller performs well, although it requires the design of several linear controllers. A look-up table technique may improve control performance, but is difficult to debug and tune. Furthermore, in complex systems where multiple inputs exist, the look-up table method may be impractical and costly to implement due to large memory requirements and the need to estimate the plant parameters on-line in the presence of severe electromagnetic interference [4,5].

Some authors have used different techniques

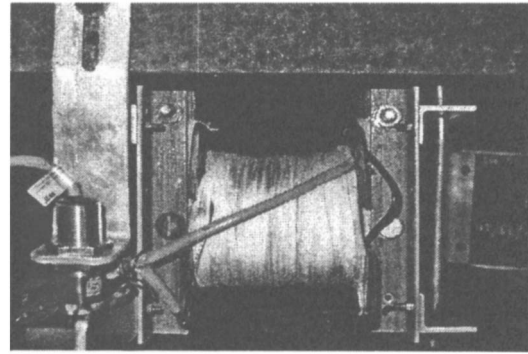


Fig. 1. Photograph of the setup during stable levitation (operating gap around 6 mm).

from the control literature (such as sliding mode control [6], μ -synthesis [7], H_∞ control [8], etc.) for controlling the large air-gap of an EMLS. All the above techniques require a digital processor for implementation and normally the order of the resulting controller is quite high. As opposed to some of the above techniques, this paper presents a control strategy, which is simple, reliable, and easily realizable through analog means. Here, the entire operating zone (4–10 mm) of the levitated system of Fig. 1 has been approximated as piecewise linear zones, and the parameters of rational linear controllers for the corresponding zones have been realized. One linear controller will act at a time for its specified zone of operation. During dynamic operation, if there is any change in the position of the levitated object (electromagnet) due to many possible reasons such as variation of load, disturbances, variation of supply voltage, etc., then the controller designed for the corresponding operating zone will be automatically switched into action instantaneously, ensuring that the stability of the levitated system is maintained. Thus an attempt has been made to increase the overall operating range of the levitated device, by ensuring automatic switching of the controllers under varying load conditions.

2. Description of the proposed scheme

The basic experimental setup (shown under levitation) is given in Fig. 1, where an electromagnet of 2.6 kg mass is suspended under a ferromagnetic guide-way (track) and may have vertical movement at different operating zones with a single degree of freedom. An inductive type position sensor is used to measure the gap between the

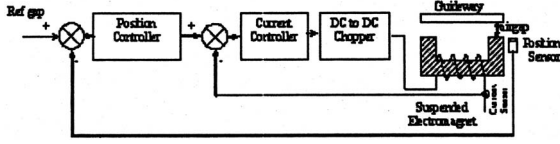


Fig. 2. Block diagram of the proposed scheme.

pole-face of the magnet and the guide-way. The block diagram of the overall control system is shown in Fig. 2. A cascade lead compensator utilizing an inner current loop and an outer position loop has been developed for stabilizing the unstable EMLS. First, the linearized models of the levitated system at different operating gaps are experimentally obtained. Output of the position loop (current reference signal) is then compared with the actual coil-current signal sensed by an LEM make (LA-55P) Hall-effect current sensor. The system parameters at different gap positions are determined experimentally.

It mainly involves measurement of the mass to be lifted, the air-gap distance, the inductance of the levitated system, and the pick up current of the magnet coil, which produces just enough attractive force to lift the mass. This experiment is repeated for different air-gap positions. For the present case, the mass of the object has been assumed to be constant. Further details of the measurement may be found in [1], where the system parameters have been determined in a similar way from the basic static characteristics (force-distance-current) of the levitated system [1]. Keeping in mind the different constraints in the lower and higher gap zones, we have selected the gap zone of 4–10 mm as the operating zone. A simple single switch based switched mode power circuit (Fig. 3) has been used for supplying the required current to the magnet coil. The circuit uses only one switch and one diode, in contrast to the two switches and two-diode configuration of the asymmetrical H-bridge circuit that is mostly reported in the lit-

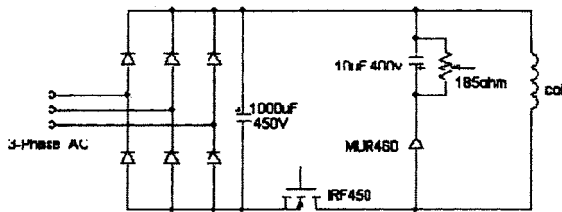


Fig. 3. Proposed power circuit.

erature [9]. In the proposed circuit there is no need for gate drive isolation as in the case of an asymmetrical bridge circuit.

In the present work, a three-step air-gap controller has been designed and fabricated. From theoretical analysis, after having an idea about the controller parameters for the three different operating zones, three sets of lead controllers were designed and fabricated, each corresponding to one zone. The controller design procedure has been described in the next section. By applying three reference-gap (position) voltages, initially some trials were made to independently tune the parameters of the controllers for the three operating gaps. In Fig. 4, the actual values of the controller parameters are given for which the prototype showed stable levitation for the corresponding reference gap settings [10–12].

3. Mathematical modeling and controller design procedure

With reference to Fig. 1 assuming that all flux generated by the electromagnet passes through the ferromagnetic guide-way and assuming that the overall magnetic circuit is linear, the instantaneous coil inductance may be expressed as

$$L(z) = \frac{N}{i(t)} \Phi_T = \frac{N^2}{R_T}, \quad (1)$$

where N is the number of turns of the coil, $i(t)$ is the instantaneous current through the coil, Φ_T is the total flux in the magnetic circuit, and R_T is the reluctance of the entire magnetic circuit.

If the reluctance of the magnetic core is assumed to be negligible when compared with the two air gaps [total length = $2z(t)$],

$$L(z) = \frac{\mu_0 N^2 A}{2z(t)}. \quad (2)$$

At any instant of time, the force of attraction between the electromagnet and the ferromagnetic rail is given by

$$F(i, z) = - \frac{d}{dz} \left[\frac{1}{2} L(z) i(t)^2 \right]. \quad (3)$$

The negative sign in the above expression indicates the direction of the generated force. The force is attractive in nature and attempts to decrease the gap length. Now putting the inductance

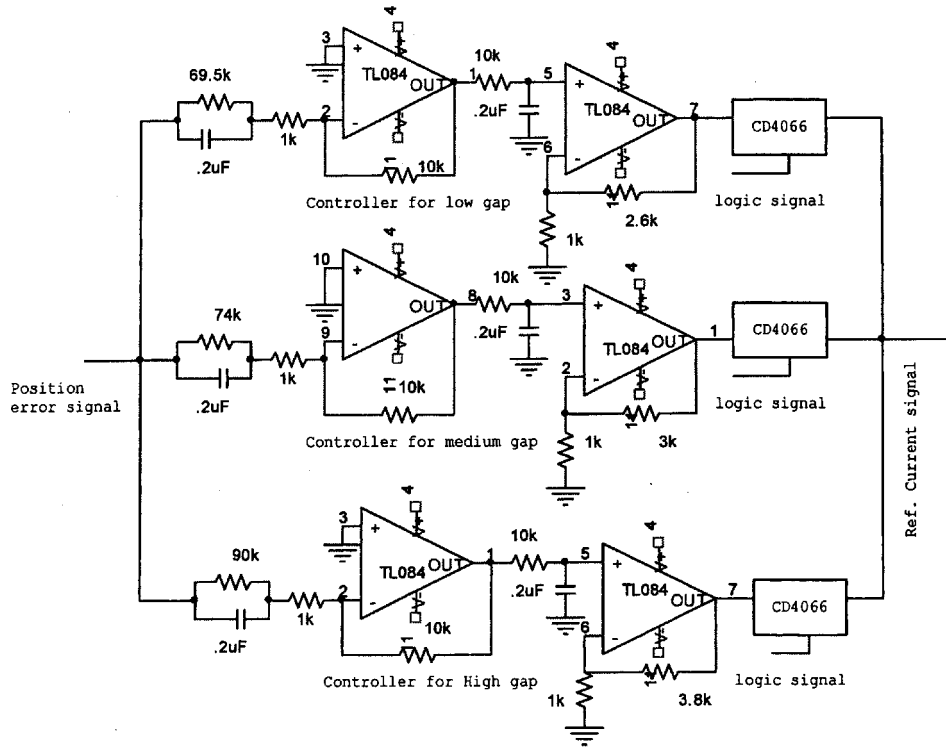


Fig. 4. Circuit diagram for three position controllers for three different zones of air gap.

value from Eq. (2) into the force Eq. (3), one can write

$$F(i, z) = \frac{\mu_0 N^2 A}{4} \left[\frac{i(t)}{z(t)} \right]^2. \quad (4)$$

So, at the equilibrium position (i_0, z_0) the normalized force equation is

$$F_0(i_0, z_0) = \frac{\mu_0 N^2 A}{4} \left[\frac{i_0}{z_0} \right]^2 = mg. \quad (5)$$

The dynamics of the electromagnet is given by the following equations:

$$m\ddot{z}(t) = -F(i, z) + mg = -\frac{\mu_0 N^2 A}{4} \left[\frac{i(t)}{z(t)} \right]^2 + mg. \quad (6)$$

The system dynamic equations are thus nonlinear and, hence, difficult to analyze. The equations are linearized about a suitable operating point (i_0, z_0) and the linearized model may be found as described below.

By neglecting second-order terms, the small signal linearized dynamic equation of the system becomes

$$\begin{aligned} m\Delta\ddot{z}(t) &= \frac{\mu_0 N^2 A}{4} \left[\frac{i_0 + \Delta i(t)}{z_0 + \Delta z(t)} \right]^2 + mg \\ &= -\frac{\mu_0 N^2 A i_0}{2z_0^2} \Delta i(t) + \frac{\mu_0 N^2 A i_0^2}{2z_0^3} \Delta z(t) \\ &= -K_a \Delta i(t) + K_z \Delta z(t). \end{aligned} \quad (7)$$

From Eq. (7), the transfer function of the plant is found as

$$G_p(s) = \frac{\Delta Z(s)}{\Delta I(s)} = -\frac{\left(\frac{K_a}{m} \right)}{\left(s^2 - \frac{K_z}{m} \right)}. \quad (8)$$

The two force constants K_a and K_z are, respectively, the slope (at the operating point) of the force-current and force-distance characteristics of the electromagnetic suspension system with mass “ m ” and can be determined experimentally.

Table 1 shows the linearized models of the levitated system at different operating air-gap positions. When the air-gap changes from 8 to 4 mm, the plant gain varies in the range (16.73–23.69) and the open-loop poles vary in the range (± 64.67)

Table 1

Linearized models of the levitated system at different operating air-gap positions.

Airgap (mm)	K_a (N/A)	K_z (N/m)	Transfer function of Plant
4	61.60	13.552×10^3	$23.69 / (s+72.19)(s-72.19)$
6	51.73	11.984×10^3	$19.89 / (s+67.89)(s-67.89)$
8	43.50	10.875×10^3	$16.73 / (s+64.67)(s-64.67)$

to ± 72.19). Thus for 100% increase in the air-gap, we note that the gain increases by 42% while the pole magnitude increases by 12%. Information about this variation of the plant parameters with air-gap is used for design and realization of corresponding controller parameters.

From Table 1, for an operating air-gap of 6 mm, the plant $G_p(s)$ has one stable pole at $s = -67.89$ and an unstable pole at $s = 67.89$. The objective is to design a phase lead compensator so that the overall closed-loop system is stabilized. The pole and zero are to be placed on the negative real axis and there are many possible pole-zero locations for stabilizing the system. The phase-lead compensator design procedure in this case is to place the zero of the compensator between 0 and -67.89 on the real axis of the s plane, while the pole of the compensator is placed about 10 times to the left of the zero position. Simulation studies are carried out to find a pole-zero and gain combination so that the system is stable and acceptable performance is obtained. It must be mentioned that since the system is inherently unstable, in general both stability and good performance cannot be achieved by using a single controller. The transfer function of the designed lead controller for an operating air-gap of 6 mm is given by

$$G_{pc}(s) = \frac{4.6(s + 64)}{(s + 500)}. \quad (9)$$

The frequency response plot of the overall closed-loop system with the lead controller is shown in Fig. 5. The compensated Bode plot (Fig. 5) of the closed-loop system shows the measures of relative stability as having a gain margin of 6.2 dB and a phase margin of 30 deg. The gain crossover frequency, which approximately determines the speed of response, is nearly 22 Hz.

The root locus plot of the overall closed-loop system with the lead controller is shown in Fig. 6. The same design technique has been utilized to

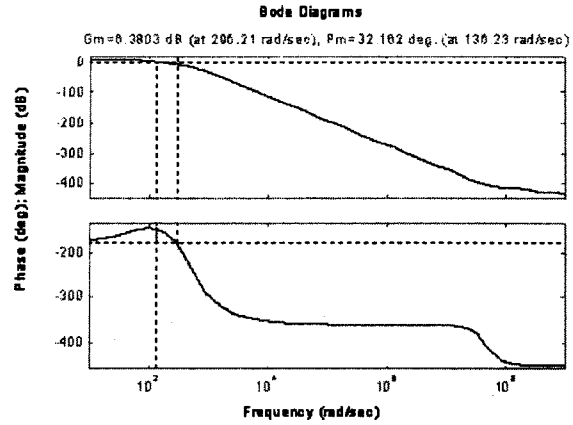


Fig. 5. Frequency response plot of the overall closed loop.

determine the controller parameters for the other operating gaps. In Table 2, the transfer functions of the controllers have been tabulated for the respective air-gap positions. Here the “pole” of the theoretical lead controller has been fixed at $s = -500$, while the position of the “zero” and the system gain are varied. For a particular operating point, simulation has been carried out to find the “zero” and gain values that give better performance in both the time and frequency domains. Based on simulation studies and theoretical designs, the hardware controller circuits have been realized while keeping provisions for varying the zero-position and the system gain. Some trials were made to obtain stable levitation for a particular air-gap and the corresponding parameters of the controller have been measured. In the practical realization, the “pole” of the compensator has been fixed at $s = -500$. Due to the simplifying assumptions made in the controller design, the actual values of the compensator network elements do not exactly match the theoretical designed values. First of all, the theoretical controller was designed based on a linearized model of the system where the parameters of the model were determined by static measurements. The actual values of the compensator network elements were measured for tuned values of the controller under stable levitation at the desired air-gap. Moreover, in a practical situation, the system parameters may vary from the theoretical values due to linear approximation of the electromagnetic force made in the theoretical design, unmodeled dynamics of the electromagnet, and other parametric uncertainties. It may be noted that with the increase of air-gap (from 4 to 8 mm), the required compensator gain

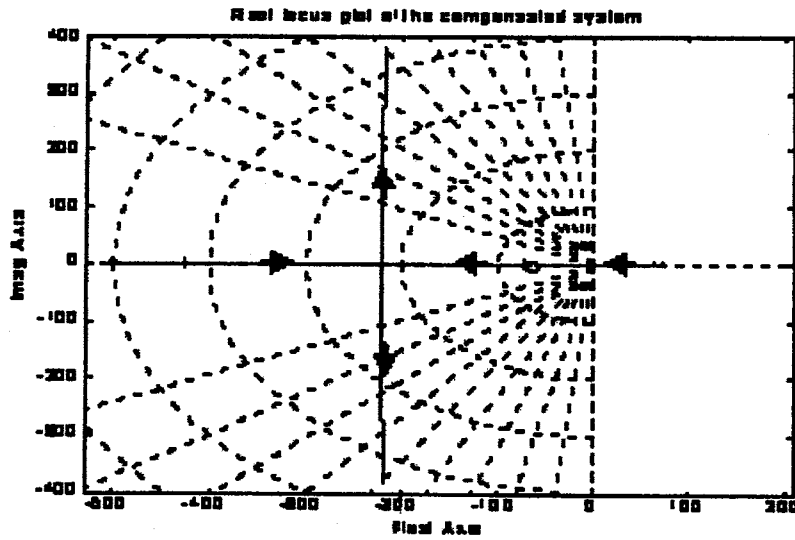


Fig. 6. Root-locus plot of the compensated system using lead compensator.

of the lead controller (both in the case of simulated and practical results) increases, since the system open-loop gain reduces with the increase of air-gap. In the s plane the compensator “zero” tends to shift toward the right (both in simulation and practice) with the increased air-gap, since the open-loop plant parameters reduce with a higher air-gap. The parameters of the controller, however, do not vary much from one air-gap to the other due to less variation of system parameters within the stipulated zone. During experimentation it has been observed that the parameters of the controller including the gain are very sensitive and may lead to instability at a particular operating point.

4. Hardware description

In order to allow smooth movement of the object through the large air gap, the gap command (reference gap) is given in the form of a ramp signal rather than a step signal. The ramp rate should correspond to the inertia of the system. For

example, to lift the magnet from its rest position when the position sensor output is, say, 6 V (corresponding to the gap distance of 10 mm) to a gap distance of 5 mm where the position-sensor output should become 1 V, the air-gap command signal is linearly changed from 6 V to 1 V in about 2.5 s. This entire range is subdivided in two or more zones. In the present case, it has been divided into three zones. Piecewise linear controllers for all these zones are realized (Fig. 4) such that when the magnet (object) is at a certain air-gap, the controller output corresponding to that particular air-gap is only chosen to decide the magnet current command. The magnet power supply circuit then responds to this current demand. The final setting of the air-gap may be different from the intermediate air-gap positions. The actual movement to the final set gap may be traversed in three steps. During each step, the chosen controller corresponds to the one designed for that particular air-gap range. Only the relevant output of the individual air-gap controller is transmitted for magnet current control through a multiplexing circuit designed using electronically controlled analog switches (IC chip 4066). Since the command signal changes rather slowly, the levitation system is always in a quasistable condition (actual air-gap follows the linear change in command distances, barring some steady-state error).

For this three-stage control, the actual position signal is simultaneously compared with two different reference air-gap voltages as shown in Fig. 7.

Table 2
Transfer functions (TF) of the controller.

Operating gap (mm)	TF of lead controller (theoretical)	TF of lead controller (practical)
4	$3.9(s+70)/(s+500)$	$2.6(s+71.94)/(s+500)$
6	$4.6(s+64)/(s+500)$	$3(s+67.56)/(s+500)$
8	$5.5(s+62)/(s+500)$	$3.8(s+55.55)/(s+500)$

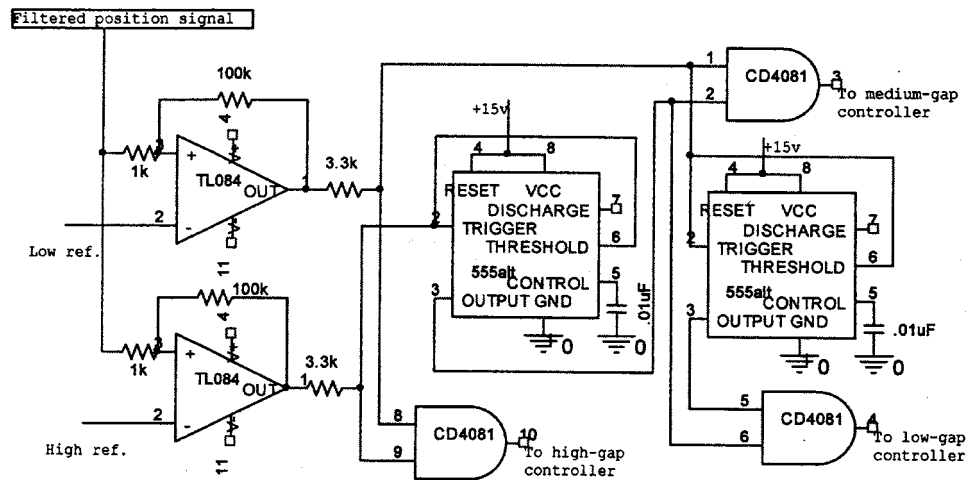


Fig. 7. Schematic diagram of the logic circuit.

When the gap-sensor voltage is larger than both the reference voltages, the controller corresponding to the highest air-gap is chosen. When the position signal lies between these two set voltages, the medium range air-gap controller is selected. When the gap-sensor voltage is lower than both the reference voltages, the third controller that corresponds to the minimum air-gap is chosen. Depending on the particular air-gap range, a particular comparator logic circuit output goes high. This logic signal then activates (closes) the corresponding analog switch. The other two switches remain open. Output points of all three analog switches are shorted and fed as a current reference signal for the magnet current controller. Only the particular switch that remains closed carries the current

command signal. Since the parameters of the current controller do not change much with a change in the air-gap, the same current controller has been used for the three different air-gaps. To decide the duty ratio of the chopper circuit, feeding current to the magnet coil, the output of the current controller is compared with a triangular wave of around 5 KHz. The comparator generates the desired PWM waveform to be used for driving the MOS-FET switch of the chopper (after required isolation and amplification).

5. Experimental results

In the oscillogram of Fig. 8, variation of actual air-gap (proportional to the distance between the

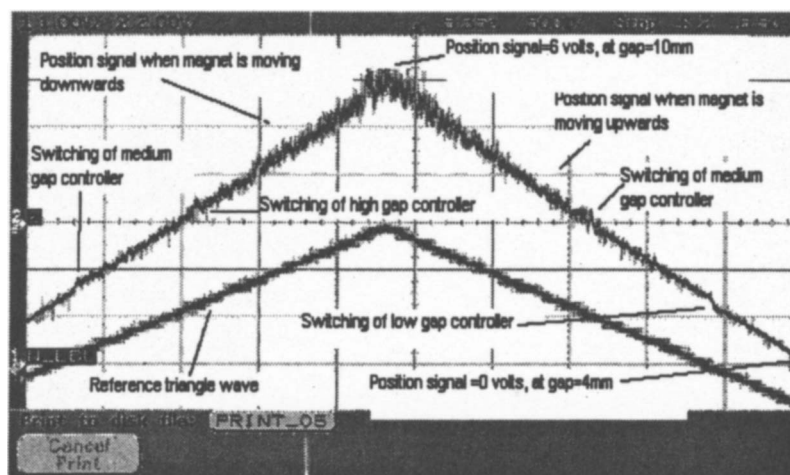


Fig. 8. Position signal (CH1) with ramp command (CH2).

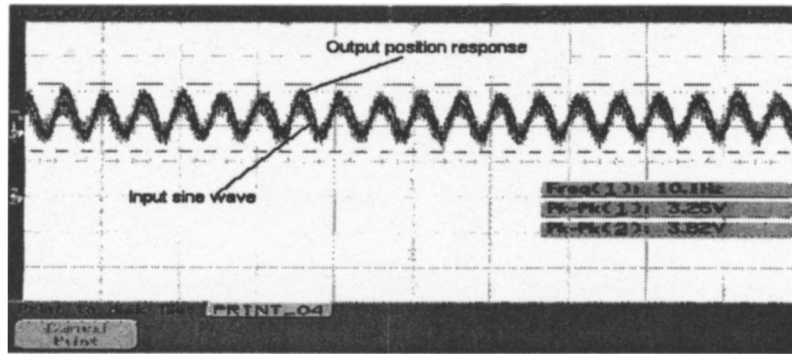


Fig. 9. Tracking performance for a sine wave (of 2 V p-p, 10 Hz).

probe surface and metallic target; 1 mm=1 V) (see CH1) with the command signal (CH2) has been recorded. A triangular wave varying between -7 V and zero (shown in channel 2) is added with a fixed signal of 8 V to form the linearly varying air-gap command signal. It is seen that, barring a small steady-state error, the electromagnet follows the slowly varying ramp command and stably covers a distance of 6 mm (from 4 to 10 mm air-gap between the magnet pole-face and the ferromagnetic guide-way). While traveling, the magnet passes through three different operating zones. The range of the operating zone is decided by the switching logic used in the circuit (Fig. 9). With the slowly increasing reference ramp command, the electromagnet moves toward a higher gap resulting in an increase of distance between the sensor and the target. The output of the position sensor (which measures the distance between the sensing surface and metallic target) is also in-

creased as recorded in CH1. When the rising edge of the position signal just crosses the low set value of reference voltage (nearly 1.4 V) of the logic circuit, the controller for the medium gap is switched on. At this instant, the operating gap of the electromagnet is nearly 5.5 mm. The existing controller works in a particular zone as long as the position signal is just below the next reference voltage (in this case, nearly 3.2 V). At this instant, the electromagnet levitates at slightly more than 7 mm gap distance. With the change of ramp command, the controller for high gap zone is automatically switched on when the position signal voltage just exceeds 3.2 V. The electromagnet is moved further toward the higher air-gap and may go up to the maximum 10 mm air-gap with this controller. During transition from one controller to the other, the continuity in the chopper command signal is disturbed, resulting in a small step change in the current command signal. The dynamics of

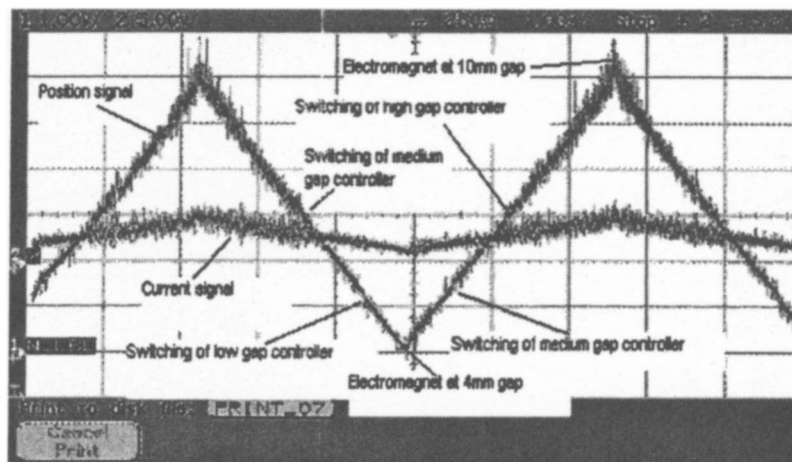


Fig. 10. Position (CH1) and current (CH2) signal for a ramp input.

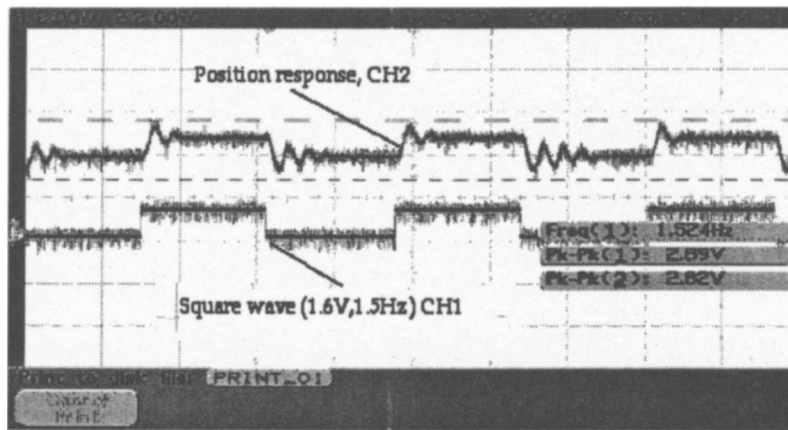


Fig. 11. Response at air-gap of 6 mm for square-wave input (1.6 V, 1.5 Hz).

the current controller, in response to this discontinuity, shows up as small transients in the actual gap signal at the switchover points and may be observed in Fig. 8. Fig. 10 shows the position signal along with the coil-current signal in response to the described reference air-gap command. In response to an increasing ramp command, as the gap between the pole-face of electromagnet and guide-way increases, the coil-current also increases to maintain the required attractive force for stable operation. This current waveform indirectly gives the position information of the electromagnet. Before implementation of the three-stage controller, the robustness and effectiveness of the individual gap controllers was tested for different kinds of disturbances (such as step, sinusoid) with changing magnitudes of air-gap and frequency of the disturbance signal. Fig. 11 shows the practical responses when subjected to a square wave disturbance (1.6 V, 1.5 Hz) at an air-gap of 6 mm. It is obvious that the closed-loop system is stable and has a satisfactory transient response. The response of a sinusoidal disturbance of up to 10 Hz, 2 V p-p shows a stable behavior (Fig. 9). However, if the frequency is increased beyond 10 Hz, the performance of the controller degrades.

6. Conclusion

The model parameters of the EMLS at different operating points have been experimentally determined and the corresponding piecewise linear switching controllers have been designed and implemented. A particular controller stabilizes the EMLS in the operating zone for which it has been

designed. This shows that the proposed method of modeling as well as controller design for the existing EMLS is practicable and successful. The practical values of the compensator network elements may not exactly match the theoretical designed values due to the simplifying assumptions made in the controller design and modeling uncertainties. Moreover, in actual practice, the system parameters may vary from the theoretical values due to linear approximation of the electromagnetic force made in the theoretical design, temperature variation, unmodeled dynamics of the electromagnet, and other parametric uncertainties.

The operating range of the electromagnet has been conclusively enhanced by using piecewise linear control. By changing the switching logic, one can vary the individual zone of operation of a controller. If the desired operating air-gap zone is larger, more linear controllers may be incorporated. In this work, the maximum operating gap has been restricted to 10 mm due to operational constraints of existing position sensor, actuator, and power amplifier.

References

- [1] Sinha, P. K., *Electromagnetic Suspension, Dynamics and Control*, Peter Peregrinus Ltd., London, 1987.
- [2] Jayawant, B. V., Review lecture on electromagnetic suspension and levitation techniques. *Proc. R. Soc. London, Ser. A* **416**, 245–320 (1988).
- [3] Trumper, D., Olson, S. M., and Subrahmanyam, P. K., Linearizing control of magnetic suspension systems. *IEEE Trans. Control Syst. Technol.* **5**(4), 427–438 (1997).
- [4] Kim, Y. C., Gain scheduled control of magnetic suspension system. *Proc. of the American Cont. Conf.*,

- pp. 3127–3131, June 1994.
- [5] Boehm, J., Gerber, R., and Kiley, N. R. C., Sensors for magnetic bearings. *IEEE Trans. Magn.* **29**(6), 2962–2964 (1993).
 - [6] Al-Muthairi, N. F. and Zribi, M., Sliding mode control of a magnetic levitation system. *Math. Probl. Eng.* **2**, 93–107 (2004).
 - [7] Lin, J. and Tho, B., Analysis and μ -based controller design for an electromagnetic suspension system. *IEEE Trans. Educ.* **41**(2), 116–129 (1998).
 - [8] Bittar, A. and Sales, R. M., H_2 and H_∞ control for Maglev vehicles. *IEEE Control Syst. Mag.* **18**(4), 18–25 (1998).
 - [9] Shirazee, N. A. and Basak, A., Electropermanent suspension system for acquiring large air-gaps to suspend loads. *IEEE Trans. Magn.* **31**(6), 4193–4195 (1995).
 - [10] Golob, M. and Tovornik, B., Modeling and control of the magnetic suspension system. *ISA Trans.* **42**, 89–100 (2003).
 - [11] Krishnan, R., *Electrical Motor Drives-Modeling, Analysis and Control*. Prentice Hall of India, New Delhi, 2003.
 - [12] Ogata, K., *Modern Control Engineering*, 3rd ed. Prentice Hall of India, New Delhi, 2000.